D3V3NDRA CLASSES

INTERNATIONAL MATHEMATICS OLYMPIAD (IMO) • TRAINING SERIES

CHAPTER 12 • DECIMALS

Grade VII (Class 7) • NCERT Advanced Curriculum Benchmark • Total Questions: 50

Candidate Name: ..... **Roll No:** ..... **School / Center:** ..... **Date:** .....

Time: 60 Mins | **Max Marks:** 100 **Total Questions:** 50 MCQs | **Marking:** +2 / -0 (No Negative) **Structure:** Sec 1 (15 LR) • Sec 2 (20 MR) • Sec 3 (10 EM) • Sec 4 (5 Achievers)

SECTION 1: LOGICAL REASONING (Q1 – Q15)

Q1. Consider the geometric sequence of decimals:

$$0.3,\ 0.09,\ 0.027,\ 0.0081,\ \ldots$$

By identifying the constant multiplication ratio connecting consecutive terms, what is the 6th term of this sequence?

(A) 0.00243 **(B)** 0.000729
(C) 0.0000729 **(D)** 0.000243

Q2. A four-digit decimal number is written in the form $N = 0.abcd$. The tenths digit a is twice the hundredths digit b, b is twice the thousandths digit c, and c is twice the ten-thousandths digit d. If $d = 1$, what is the numerical value of N?

(A) 0.4218 **(B)** 0.8241
(C) 0.8421 **(D)** 0.1248

Q3. In a 3×3 decimal magic addition square, the sum along every row, column, and diagonal is strictly equal to 1.5. If the center cell contains 0.5 and the top-left cell contains 0.8, what decimal must occupy the bottom-right cell?

(A) 0.3 **(B)** 0.4
(C) 0.1 **(D)** 0.2

Q4. An operation $\diamond$ on decimals is defined by the algebraic rule $x \diamond y = x \times y + 0.1 \times (x + y)$. Evaluate the numerical value of $0.4 \diamond 0.5$.

(A) 0.20 **(B)** 0.29
(C) 0.31 **(D)** 0.25

Q5. What is the missing decimal in the alternating pattern sequence:

$$1.2,\ 2.4,\ 2.8,\ 5.6,\ 6.0,\ 12.0,\ [\ ?\ ]$$

(A) 14.0 **(B)** 24.0
(C) 12.8 **(D)** 12.4

Q6. Which of the following four fractions converts into a NON-TERMINATING recurring decimal when expanded?

(A) $\frac{13}{40}$ **(B)** $\frac{7}{125}$
(C) $\frac{11}{75}$ **(D)** $\frac{9}{120}$

Q7. On a balanced two-pan scale, 4 identical steel blocks each of mass $0.75\ \text{kg}$ and 3 wooden spheres balance 2 identical steel blocks of mass $0.75\ \text{kg}$ and 7 wooden spheres. What is the mass of ONE wooden sphere?

(A) $0.375\ \text{kg}$ **(B)** $0.25\ \text{kg}$
(C) $0.50\ \text{kg}$ **(D)** $0.125\ \text{kg}$

Q8. Four unknown decimal quantities w, x, y, z satisfy the equality chain:

$$w + 0.4 = x - 0.2 = y + 0.7 = z - 0.5$$

Which of the four variables has the STRICTLY LARGEST numerical value?

(A) w **(B)** z
(C) x **(D)** y

Q9. How many single digits $d \in \{0, 1, 2, \ldots, 9\}$ satisfy the strict double inequality:

$$0.345 < 0.3d2 < 0.381$$

(A) 3 **(B)** 4
(C) 5 **(D)** 6

Q10. If the decimal cryptarithm $0.A + 0.0A + 0.00A = 0.777$ holds true, where A represents a single non-zero digit, what is the numerical value of the product $A \times 0.A$?

(A) 0.49 **(B)** 4.9
(C) 49.0 **(D)** 0.049

Q11. A secret number N is multiplied by 100, then divided by 10, and finally multiplied by 0.01. If the resulting value is 0.0345, what was the original number N?

(A) 3.45 **(B)** 0.0345
(C) 0.345 **(D)** 0.00345

Q12. Which of the following decimal multiplication products has a DIFFERENT total count of decimal places prior to simplifying any trailing zeros?

(A) 0.05×0.16 **(B)** 0.008×0.1
(C) 0.4×0.025 **(D)** 0.2×0.04

Q13. In a 2×2 grid with decimal entries $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$, the cross-metric value is defined as $(a \times d - b \times c)$. If $a = 0.6, d = 0.5, b = 0.4$, and $c = 0.3$, what is the cross-metric value?

(A) 0.22 **(B)** 0.18
(C) 0.42 **(D)** 0.12

Q14. Evaluate the exact sum of the first four terms of the decimal geometric series:

$$S_4 = 0.9 + 0.09 + 0.009 + 0.0009$$

(A) 0.999 **(B)** 1.0
(C) 0.9099 **(D)** 0.9999

Q15. For any positive decimal x strictly between 0 and 1 (for example, $x = 0.1$), which of the following expressions is GUARANTEED to have the GREATEST numerical value?

(A) $\frac{1}{x}$ **(B)** x
(C) $\sqrt{x}$ **(D)** x^2

SECTION 2: MATHEMATICAL REASONING (Q16 – Q35)

Q16. Evaluate the multi-step arithmetic expression:

$$2.5 \times [1.8 - 0.3 \times (4.2 - 1.2)] \div 0.05$$

(A) 4.5 **(B)** 450
(C) 45 **(D)** 0.45

Q17. Evaluate the sum of the decimal quotients:

$$\frac{0.00072}{0.00009} + \frac{0.0048}{0.00012}$$

(A) 48 **(B)** 4.8
(C) 480 **(D)** 8.4

Q18. Evaluate the algebraic expression with decimals using factoring:

$$\frac{0.87 \times 0.87 - 0.13 \times 0.13}{0.87 - 0.13}$$

(A) 1.0 **(B)** 0.74
(C) 1.74 **(D)** 0.87

Q19. Evaluate the ratio of cubic sums:

$$\frac{0.3 \times 0.3 \times 0.3 + 0.06 \times 0.06 \times 0.06}{0.6 \times 0.6 \times 0.6 + 0.12 \times 0.12 \times 0.12}$$

(A) 0.25 **(B)** 0.125
(C) 0.5 **(D)** 0.0625

Q20. When the mixed recurring decimal $0.2\overline{7}$ is converted into an irreducible fraction $\frac{p}{q}$, what is the value of $(p + q)$?

(A) 25 **(B)** 27
(C) 23 **(D)** 29

Q21. Evaluate the sum and difference of recurring decimals:

$$0.\bar{6} + 0.\bar{3} - 0.\bar{2}$$

Express the result as a recurring decimal.

(A) $0.\bar{8}$ **(B)** 0.75
(C) $0.\bar{6}$ **(D)** $0.\bar{7}$

Q22. The fraction $\frac{13}{2^2 \times 5^4}$ is converted into its exact decimal representation. After how many decimal places will the decimal terminate?

(A) 2 **(B)** 6
(C) 8 **(D)** 4

Q23. Evaluate the difference of squares:

$$(3.65)^2 - (1.35)^2$$

(A) 10.5 **(B)** 12.5
(C) 11.5 **(D)** 11.25

Q24. A rectangular garden has length $12.5\ \text{m}$ and width $8.4\ \text{m}$. A uniform concrete pathway of width $1.2\ \text{m}$ is constructed all around the outside of the garden. What is the total area of the pathway?

(A) $55.92\ \text{m}^2$ **(B)** $50.16\ \text{m}^2$
(C) $52.48\ \text{m}^2$ **(D)** $60.24\ \text{m}^2$

Q25. Solve for the decimal unknown x in the linear equation:

$$0.3(2x - 1.5) = 0.15(x + 4.5)$$

(A) 3.0 **(B)** 1.5
(C) 2.0 **(D)** 2.5

Q26. If $a = 0.5$, what is the numerical value of the algebraic fraction:

$$\frac{a^2 + a}{a^3 - a}$$

(A) 2.0 **(B)** -0.5
(C) 1.0 **(D)** -2.0

Q27. For which integer exponent k does the product 0.0034×10^k lie strictly between 30 and 40?

(A) 3 **(B)** 4
(C) 5 **(D)** 2

Q28. A student attempts to evaluate $4.5 \div 0.09$. In Step 1, he multiplies dividend and divisor by 10 to get $45 \div 0.9$. In Step 2, he divides $45 \div 9$ to get 5. In Step 3, he states the final answer is 0.5. Which step contains the fundamental mathematical error?

(A) Step 1 **(B)** Step 3
(C) Step 2 **(D)** There is no error

Q29. Evaluate the value of the algebraic decimal expression:

$$\frac{0.064 + 0.027}{0.16 - 0.12 + 0.09}$$

(A) 0.7 **(B)** 0.07
(C) 7.0 **(D)** 0.49

Q30. Express the metric sum entirely in metres as a single decimal:

$$4\ \text{km}\ 25\ \text{m} + 350\ \text{cm} + 850\ \text{mm}$$

(A) $4029.35\ \text{m}$ **(B)** $4028.5\ \text{m}$
(C) $4254.35\ \text{m}$ **(D)** $4025.435\ \text{m}$

Q31. Which of the following lists the decimals in strictly DESCENDING order (from largest to smallest)?

(A) $0.755,\ 0.75,\ 0.705,\ 0.075$ **(B)** $0.75,\ 0.755,\ 0.705,\ 0.075$
(C) $0.755,\ 0.705,\ 0.75,\ 0.075$ **(D)** $0.075,\ 0.705,\ 0.75,\ 0.755$

Q32. Evaluate the product:

$$0.1 \times 0.01 \times 0.001 \times 10^5$$

(A) 0.01 **(B)** 1.0
(C) 0.1 **(D)** 0.001

Q33. Evaluate the complex quotient of quotients:

$$\frac{1.44 \div 0.12}{0.036 \div 0.003}$$

(A) 1.0 **(B)** 10.0
(C) 0.1 **(D)** 1.2

Q34. Evaluate the difference of recurring decimals as an irreducible fraction:

$$0.\overline{72} - 0.\overline{27}$$

(A) $\frac{9}{22}$ **(B)** $\frac{5}{11}$
(C) $\frac{45}{99}$ **(D)** $\frac{1}{2}$

Q35. A measured length rounded to one decimal place is given as $8.4\ \text{cm}$. Which of the following intervals represents the complete set of possible true values x of this measurement?

(A) $8.30 \le x \le 8.50$ **(B)** $8.35 \le x < 8.45$
(C) $8.35 < x \le 8.45$ **(D)** $8.39 \le x \le 8.41$

SECTION 3: EVERYDAY MATHEMATICS (Q36 – Q45)

Q36. A family car travels a total distance of $456.8\ \text{km}$ on a full tank containing $28.55\ \text{litres}$ of petrol. What is the vehicle's fuel economy in kilometres per litre?

(A) $15.5\ \text{km/L}$ **(B)** $16.5\ \text{km/L}$
(C) $14.8\ \text{km/L}$ **(D)** $16.0\ \text{km/L}$

Q37. At a supermarket, Ananya purchased $3.75\ \text{kg}$ of apples at $\text{Rs}\ 120.40$ per kg, and $2.5\ \text{kg}$ of grapes at $\text{Rs}\ 85.60$ per kg. What was her total expenditure?

(A) $\text{Rs}\ 665.50$ **(B)** $\text{Rs}\ 650.00$
(C) $\text{Rs}\ 672.25$ **(D)** $\text{Rs}\ 658.75$

Q38. An international exchange bank quotes the foreign exchange spot rate of $1\ \text{US Dollar} = \text{Rs}\ 83.4520$. How many Indian Rupees will an overseas student receive upon exchanging 250 USD?

(A) $\text{Rs}\ 20,863.00$ **(B)** $\text{Rs}\ 20,860.00$
(C) $\text{Rs}\ 20,865.50$ **(D)** $\text{Rs}\ 20,850.00$

Q39. A pediatric medicine requires a dosage of $0.075\ \text{mL}$ of active liquid suspension per kilogram of body weight per day. For a child weighing $24.8\ \text{kg}$, what is the exact daily volume of medication required?

(A) $1.85\ \text{mL}$ **(B)** $1.92\ \text{mL}$
(C) $1.75\ \text{mL}$ **(D)** $1.86\ \text{mL}$

Q40. In an Olympic 100-metre final, Runner A clocked 10.145 seconds, Runner B clocked 10.182 seconds, and Runner C clocked 10.098 seconds. What is the time difference between the gold medalist and the bronze medalist?

(A) 0.047 seconds **(B)** 0.037 seconds
(C) 0.084 seconds **(D)** 0.096 seconds

Q41. A city meteorological station recorded a rainfall of $35.4\ \text{mm}$ during a thunderstorm. If the flat rooftop of an apartment complex has a surface area of $120.5\ \text{m}^2$, how many litres of rainwater fell on the rooftop? ($1\ \text{m}^3 = 1000$ litres)

(A) 426.57 litres **(B)** $4,265.7$ litres
(C) $42,657$ litres **(D)** $3,540.0$ litres

Q42. An electricity utility charges a flat meter maintenance fee of $\text{Rs}\ 85.50$ plus $\text{Rs}\ 6.75$ per unit of electricity consumed. If a household consumed 245.8 units in October, what was the total billing amount?

(A) $\text{Rs}\ 1,659.15$ **(B)** $\text{Rs}\ 1,740.00$
(C) $\text{Rs}\ 1,750.50$ **(D)** $\text{Rs}\ 1,744.65$

Q43. A factory spool carries a copper grounding wire measuring 157.5 metres. How many identical grounding leads of length 1.75 metres can be cut from this spool?

(A) 85 **(B)** 95
(C) 90 **(D)** 100

Q44. A certified 22-karat gold commemorative medallion weighs 11.664 grams and contains 0.916 pure gold by mass. What is the mass of pure gold contained in the medallion rounded to three decimal places?

(A) $10.500\ \text{g}$ **(B)** $10.725\ \text{g}$
(C) $10.684\ \text{g}$ **(D)** $11.000\ \text{g}$

Q45. A cyclist rides at a constant cruising speed of $18.6\ \text{km/h}$ for a duration of 2.75 hours. What is the total distance covered by the cyclist?

(A) $51.15\ \text{km}$ **(B)** $50.25\ \text{km}$
(C) $52.00\ \text{km}$ **(D)** $49.85\ \text{km}$

SECTION 4: ACHIEVERS SECTION / HOTS (Q46 – Q50)

Q46. Evaluate the value of the finite telescoping product of four decimal factors:

$$P = (1 - 0.5) \times (1 - 0.\bar{3}) \times (1 - 0.25) \times (1 - 0.2)$$

Express the final result as a single decimal.

(A) 0.1 **(B)** 0.2
(C) 0.25 **(D)** 0.05

Q47. In the recurring decimal expansion of the unit fraction $\frac{1}{7} = 0.\overline{142857}$, what is the 2026th digit to the right of the decimal point?

(A) 4 **(B)** 2
(C) 8 **(D)** 5

Q48. The nine decimals $0.1, 0.2, 0.3, \ldots, 0.9$ are written on a chalkboard. In each move, a student erases any two numbers x and y and replaces them with the single value $(x + y - x \times y)$. This step is repeated 8 times until only one number F remains. What is the value of $(1 - F)$ expressed in scientific notation?

(A) 3.6288×10^{-5} **(B)** 3.6288×10^{-4}
(C) 3.6288×10^{-3} **(D)** 1.0×10^{-4}

Q49. For any two positive decimals x and y satisfying the linear constraint $x + y = 1.6$, what is the ABSOLUTE MAXIMUM possible numerical value achievable by the product $x \times y$?

(A) 0.60 **(B)** 0.64
(C) 0.68 **(D)** 0.56

Q50. If the continued fraction with recurring decimals is evaluated as an irreducible rational fraction $\frac{p}{q}$:

$$x = 0.\bar{6} + \cfrac{1}{1 + \frac{1}{0.\bar{3}}}$$

What is the numerical value of the sum $(p + q)$?

(A) 19 **(B)** 25
(C) 21 **(D)** 23

ANSWER KEY & DETAILED STEP-BY-STEP SOLUTIONS

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QUICK ANSWER KEY MATRIX (50 MCQS)

1	2	3	4	5	6	7	8	9	10
B	**C**	**D**	**B**	**D**	**C**	**A**	**B**	**A**	**B**
11	12	13	14	15	16	17	18	19	20
C	**D**	**B**	**D**	**A**	**C**	**A**	**A**	**B**	**C**
21	22	23	24	25	26	27	28	29	30
D	**D**	**C**	**A**	**D**	**D**	**B**	**C**	**A**	**A**
31	32	33	34	35	36	37	38	39	40
A	**C**	**A**	**B**	**B**	**D**	**A**	**A**	**D**	**C**
41	42	43	44	45	46	47	48	49	50
B	**D**	**C**	**C**	**A**	**B**	**C**	**B**	**B**	**D**

COMPREHENSIVE STEP-BY-STEP SOLUTIONS & CONCEPTUAL FOUNDATIONS

Q1 [LOGICAL REASONING] **Correct Option: (B)** 0.000729

Geometric Decimal Sequence:
Notice that each term is obtained by multiplying the previous term by 0.3:
- $T_1 = 0.3 = (0.3)^1$
- $T_2 = 0.09 = (0.3)^2$
- $T_3 = 0.027 = (0.3)^3$
- $T_4 = 0.0081 = (0.3)^4$
- $T_5 = (0.3)^5 = 0.00243$
- $T_6 = (0.3)^6 = 0.000729$

Each power increases the count of decimal places by 1. For $(0.3)^6$, $3^6 = 729$ with 6 decimal places, giving 0.000729.

Q2 [LOGICAL REASONING] **Correct Option: (C)** 0.8421

Decimal Place Value Deduction:
Given $d = 1$:
- $c = 2d = 2(1) = 2$
- $b = 2c = 2(2) = 4$
- $a = 2b = 2(4) = 8$

Substituting these digits into the place values:

$$N = 0.abcd = 0.8421$$

Q3 [LOGICAL REASONING] **Correct Option: (D)** 0.2

Magic Square Invariant:
The main diagonal containing top-left (TL), center (C), and bottom-right (BR) must sum to the constant magic sum $S = 1.5$:

$$TL + C + BR = 1.5$$

Substituting $TL = 0.8$ and $C = 0.5$:

$$0.8 + 0.5 + BR = 1.5 \implies 1.3 + BR = 1.5 \implies BR = 1.5 - 1.3 = 0.2$$

Q4 [LOGICAL REASONING] **Correct Option: (B)** 0.29

Relational Operational Logic:
1. Product term: $x \times y = 0.4 \times 0.5 = 0.20$.
2. Sum term: $0.1 \times (x + y) = 0.1 \times (0.4 + 0.5) = 0.1 \times 0.9 = 0.09$.
3. Combine:

$$0.4 \diamond 0.5 = 0.20 + 0.09 = 0.29$$

Q5 [LOGICAL REASONING] **Correct Option: (D)** 12.4

Alternating Pattern Deduction:
Examine the operational steps between consecutive terms:
- $1.2 \times 2 = 2.4$
- $2.4 + 0.4 = 2.8$
- $2.8 \times 2 = 5.6$
- $5.6 + 0.4 = 6.0$
- $6.0 \times 2 = 12.0$

The alternating rule is: multiply by 2, then add 0.4.
The next step is to add 0.4:

$$12.0 + 0.4 = 12.4$$

Q6 [LOGICAL REASONING] **Correct Option: (C)** $\frac{11}{75}$

Terminating Decimal Condition:
A reduced fraction $\frac{p}{q}$ yields a terminating decimal if and only if the prime factorization of denominator q contains only 2 and/or 5.
- $\frac{13}{40}$: $40 = 2^3 \times 5$ (Terminating: 0.325)
- $\frac{7}{125}$: $125 = 5^3$ (Terminating: 0.056)
- $\frac{9}{120} = \frac{3}{40}$: $40 = 2^3 \times 5$ (Terminating: 0.075)
- $\frac{11}{75}$: $75 = 3 \times 5^2$. The prime factor 3 in the denominator cannot be canceled, producing a non-terminating recurring decimal ($0.14666\ldots$).

Q7 [LOGICAL REASONING] **Correct Option: (A)** 0.375 kg

Balance Scale Logical Deduction:
Let $B = 0.75$ kg and s be the mass of one wooden sphere.
The balance equation is:

$$4B + 3s = 2B + 7s$$

Subtract $2B + 3s$ from both sides:

$$2B = 4s \implies s = \frac{2B}{4} = \frac{B}{2}$$

Substitute $B = 0.75$ kg:

$$s = \frac{0.75}{2} = 0.375 \text{ kg}$$

Q8 [LOGICAL REASONING] **Correct Option: (B)** z

Relative Magnitude Logic:
Let the common constant value of the chain be K:
- $w = K - 0.4$
- $x = K + 0.2$
- $y = K - 0.7$
- $z = K + 0.5$

Ordering the variables from smallest to largest:

$$y < w < x < z$$

Since $K + 0.5$ is strictly greater than all other expressions, z has the largest numerical value.

Q9 [LOGICAL REASONING] **Correct Option: (A)** 3

Decimal Digit Bounding:
Compare place by place:
- For $d \leq 4$: $0.342 < 0.345$, which fails the lower bound.
- For $d = 5$: $0.345 < 0.352 < 0.381$ (Valid).
- For $d = 6$: $0.345 < 0.362 < 0.381$ (Valid).
- For $d = 7$: $0.345 < 0.372 < 0.381$ (Valid).
- For $d = 8$: $0.382 > 0.381$, which exceeds the upper bound.

The digits satisfying the inequality are $d \in \{5, 6, 7\}$, giving exactly 3 digits.

Q10 [LOGICAL REASONING] **Correct Option: (B)** 4.9

Decimal Cryptarithm Deduction:
Combine the terms:

$$0.A + 0.0A + 0.00A = A \times (0.1 + 0.01 + 0.001) = A \times 0.111 = 0.AAA$$

Given $0.AAA = 0.777$, we directly deduce $A = 7$.
Now compute the required product:

$$A \times 0.A = 7 \times 0.7 = 4.9$$

Q11 [LOGICAL REASONING] **Correct Option: (C)** 0.345

Directional Power of 10 Shifts:
Trace the net transformation on N:

$$\text{Result} = N \times 100 \div 10 \times 0.01 = N \times 10 \times 0.01 = N \times 0.1$$

We are given:

$$N \times 0.1 = 0.0345 \implies N = \frac{0.0345}{0.1} = 0.0345 \times 10 = 0.345$$

Q12 [LOGICAL REASONING] **Correct Option: (D)** 0.2×0.04

Counting Decimal Places:
Count the sum of decimal places of the two factors:
• Option A: 0.2 (1 place) $\times 0.04$ (2 places) $\implies 1+2=3$ decimal places.
• Option B: 0.05 (2 places) $\times 0.16$ (2 places) $\implies 2+2=4$ decimal places.
• Option C: 0.008 (3 places) $\times 0.1$ (1 place) $\implies 3+1=4$ decimal places.
• Option D: 0.4 (1 place) $\times 0.025$ (3 places) $\implies 1+3=4$ decimal places.
Option A has 3 decimal places, whereas B, C, and D all have 4 decimal places.

Q13 [LOGICAL REASONING] **Correct Option: (B)** 0.18

Decimal Cross-Metric Calculation:
1. Main diagonal product: $a \times d = 0.6 \times 0.5 = 0.30$.
2. Anti-diagonal product: $b \times c = 0.4 \times 0.3 = 0.12$.
3. Difference:

$$ad - bc = 0.30 - 0.12 = 0.18$$

Q14 [LOGICAL REASONING] **Correct Option: (D)** 0.9999

Decimal Expansion Sum:
Align by place value:
\begin{array}{r@{\quad}l} & 0.9000 \\ + & 0.0900 \\ + & 0.0090 \\ + & 0.0009 \\ \hline & 0.9999 \end{array}

Q15 [LOGICAL REASONING] **Correct Option: (A)** $\frac{1}{x}$

Ordering Powers and Inverses for $0 < x < 1$:
When $0 < x < 1$ (e.g. $x = 0.1$):
• $x^2 = 0.01$
• $x = 0.1$
• $\sqrt{x} = \sqrt{0.1} \approx 0.316$
• $\frac{1}{x} = \frac{1}{0.1} = 10$
The strict inequality chain is:

$$x^2 < x < \sqrt{x} < 1 < \frac{1}{x}$$

Thus, $\frac{1}{x}$ is always the greatest.

Q16 [MATHEMATICAL REASONING] **Correct Option: (C)** 45

BODMAS Order of Operations with Decimals:
1. Innermost brackets: $4.2 - 1.2 = 3.0$
2. Multiplication inside brackets: $0.3 \times 3.0 = 0.9$
3. Subtraction inside brackets: $1.8 - 0.9 = 0.9$
4. Multiplication: $2.5 \times 0.9 = 2.25$
5. Division:

$$\frac{2.25}{0.05} = \frac{225}{5} = 45$$

Q17 [MATHEMATICAL REASONING] **Correct Option: (A)** 48

Division of Decimals via Scaling:
1. First quotient: Multiply numerator and denominator by 10^5:

$$\frac{0.00072 \times 10^5}{0.00009 \times 10^5} = \frac{72}{9} = 8$$

2. Second quotient: Multiply numerator and denominator by 10^5:

$$\frac{0.0048 \times 10^5}{0.00012 \times 10^5} = \frac{480}{12} = 40$$

3. Sum:

$$8 + 40 = 48$$

Q18 [MATHEMATICAL REASONING] **Correct Option: (A)** 1.0

Difference of Squares Factoring:
Use the algebraic identity $a^2 - b^2 = (a-b)(a+b)$ with $a = 0.87$ and $b = 0.13$:

$$\frac{a^2 - b^2}{a - b} = \frac{(a-b)(a+b)}{a-b} = a + b$$

Substitute values:

$$0.87 + 0.13 = 1.0$$

Q19 [MATHEMATICAL REASONING] **Correct Option: (B)** 0.125

Scaling Factor in Cubic Expressions:
Notice that in the denominator:

$$0.6 = 2 \times 0.3 \implies (0.6)^3 = 2^3 \times (0.3)^3 = 8 \times (0.3)^3$$
$$0.12 = 2 \times 0.06 \implies (0.12)^3 = 2^3 \times (0.06)^3 = 8 \times (0.06)^3$$

Factor 8 out of the denominator:

$$\text{Denominator} = 8 \times \left((0.3)^3 + (0.06)^3\right)$$

Therefore:

$$\text{Ratio} = \frac{(0.3)^3 + (0.06)^3}{8\left((0.3)^3 + (0.06)^3\right)} = \frac{1}{8} = 0.125$$

Q20 [MATHEMATICAL REASONING] **Correct Option: (C)** 23

Converting Mixed Recurring Decimals:
Let $x = 0.2\overline{7} = 0.2777\ldots$
Multiply by 10: $10x = 2.777\ldots$
Multiply by 100: $100x = 27.777\ldots$
Subtract:

$$100x - 10x = 27.777\cdots - 2.777\cdots \implies 90x = 25 \implies x = \frac{25}{90}$$

Simplify to lowest terms by dividing numerator and denominator by 5:

$$x = \frac{5}{18}$$

Thus, $p = 5$ and $q = 18$ with $\gcd(5, 18) = 1$.

$$p + q = 5 + 18 = 23$$

Q21 [MATHEMATICAL REASONING] **Correct Option: (D)** $0.\bar{7}$

Recurring Decimal Arithmetic:
Convert to rational fractions:
• $0.\bar{6} = \frac{6}{9} = \frac{2}{3}$
• $0.\bar{3} = \frac{3}{9} = \frac{1}{3}$
• $0.\bar{2} = \frac{2}{9}$
Evaluate:

$$\frac{2}{3} + \frac{1}{3} - \frac{2}{9} = 1 - \frac{2}{9} = \frac{7}{9} = 0.\bar{7}$$

Q22 [MATHEMATICAL REASONING] **Correct Option: (D)** 4

Termination Decimal Criterion:
The number of decimal places before termination for a fraction $\frac{p}{2^a \times 5^b}$ (in lowest terms) is equal to $\max(a, b)$.
Here, $a = 2$ and $b = 4$.

$$\max(2, 4) = 4 \text{ decimal places}$$

Verification:

$$\frac{13}{4 \times 625} = \frac{13}{2500} = \frac{13 \times 4}{10000} = \frac{52}{10000} = 0.0052 \text{ (exactly 4 places)}$$

Q23 [MATHEMATICAL REASONING] **Correct Option: (C)** 11.5

Difference of Squares Evaluation:
Use $a^2 - b^2 = (a-b)(a+b)$:
• $a - b = 3.65 - 1.35 = 2.30$
• $a + b = 3.65 + 1.35 = 5.00$
Multiply:

$$(3.65)^2 - (1.35)^2 = 2.30 \times 5.00 = 11.5$$

Q24 [MATHEMATICAL REASONING] **Correct Option: (A)** $55.92\ \text{m}^2$

Pathway Area with Decimals:
1. Inner garden area:

$$A_{\text{inner}} = 12.5 \times 8.4 = 105.00\ \text{m}^2$$

2. Outer dimensions with 1.2 m border on both sides:

$$L_{\text{outer}} = 12.5 + 2(1.2) = 12.5 + 2.4 = 14.9\ \text{m}$$
$$W_{\text{outer}} = 8.4 + 2(1.2) = 8.4 + 2.4 = 10.8\ \text{m}$$

3. Outer area:

$$A_{\text{outer}} = 14.9 \times 10.8 = 160.92\ \text{m}^2$$

4. Pathway area:

$$A_{\text{path}} = 160.92 - 105.00 = 55.92\ \text{m}^2$$

Q25 [MATHEMATICAL REASONING] **Correct Option: (D)** 2.5

Solving Linear Equations with Decimals:
Divide both sides by 0.15:

$$\frac{0.3}{0.15}(2x - 1.5) = x + 4.5 \implies 2(2x - 1.5) = x + 4.5$$

Expand brackets:

$$4x - 3.0 = x + 4.5$$

Transpose terms:

$$4x - x = 4.5 + 3.0 \implies 3x = 7.5 \implies x = \frac{7.5}{3} = 2.5$$

Q26 [MATHEMATICAL REASONING] **Correct Option: (D)** -2.0

Algebraic Factoring with Decimals:
Factor the numerator and denominator:

$$\frac{a(a+1)}{a(a^2-1)} = \frac{a(a+1)}{a(a-1)(a+1)}$$

For $a \neq 0$ and $a \neq -1$, cancel common factors $a(a+1)$:

$$= \frac{1}{a-1}$$

Substitute $a = 0.5$:

$$\frac{1}{0.5-1} = \frac{1}{-0.5} = -2.0$$

Q27 [MATHEMATICAL REASONING] **Correct Option: (B)** 4

Powers of 10 Decimal Shift:
Test integer values of k:
• $k = 2 : 0.0034 \times 100 = 0.34$
• $k = 3 : 0.0034 \times 1000 = 3.4$
• $k = 4 : 0.0034 \times 10000 = 34.0$
Since $30 < 34.0 < 40$, the required exponent is $k = 4$.

Q28 [MATHEMATICAL REASONING] **Correct Option: (C) Step 2**

Error Analysis in Decimal Division:
Correct evaluation:

$$4.5 \div 0.09 = \frac{4.5 \times 100}{0.09 \times 100} = \frac{450}{9} = 50$$

In Step 1, he multiplied by 10, which made the divisor 0.9 (still a decimal). Then in Step 2, he ignored the decimal point in 0.9 and treated it as 9, computing $45 \div 9 = 5$ instead of $45 \div 0.9 = 50$. Hence Step 2 contains the primary error.

Q29 [MATHEMATICAL REASONING] **Correct Option: (A)** 0.7

Sum of Cubes Identity with Decimals:
Notice that:

$$0.064 = (0.4)^3, \quad 0.027 = (0.3)^3$$
$$0.16 = (0.4)^2, \quad 0.12 = 0.4 \times 0.3, \quad 0.09 = (0.3)^2$$

Using the algebraic identity $\frac{a^3+b^3}{a^2-ab+b^2} = a + b$ with $a = 0.4$ and $b = 0.3$:

$$= 0.4 + 0.3 = 0.7$$

Q30 [MATHEMATICAL REASONING] **Correct Option: (A)** $4029.35\ \text{m}$

Metric Unit Decimal Conversion:
Convert each term to metres:
• $4\ \text{km}\ 25\ \text{m} = 4000 + 25 = 4025.00\ \text{m}$
• $350\ \text{cm} = \frac{350}{100} = 3.50\ \text{m}$
• $850\ \text{mm} = \frac{850}{1000} = 0.85\ \text{m}$
Sum:

$$4025.00 + 3.50 + 0.85 = 4029.35\ \text{m}$$

Q31 [MATHEMATICAL REASONING] **Correct Option: (A)** $0.755,\ 0.75,\ 0.705,\ 0.075$

Ordering Decimals:
Normalize all numbers to three decimal places:
• $0.755 = 0.755$
• $0.75 = 0.750$
• $0.705 = 0.705$
• $0.075 = 0.075$
Comparing thousandths: $755 > 750 > 705 > 75$.
Descending order: $0.755, 0.75, 0.705, 0.075$.

Q32 [MATHEMATICAL REASONING] **Correct Option: (C)** 0.1

Index Notation for Decimal Products:
Write in powers of 10:

$$10^{-1} \times 10^{-2} \times 10^{-3} \times 10^{5} = 10^{-1-2-3+5} = 10^{-6+5} = 10^{-1} = 0.1$$

Q33 [MATHEMATICAL REASONING] **Correct Option: (A)** 1.0

Quotient Simplification:
1. Numerator: $1.44 \div 0.12 = \frac{144}{12} = 12$
2. Denominator: $0.036 \div 0.003 = \frac{36}{3} = 12$
3. Ratio:

$$\frac{12}{12} = 1.0$$

Q34 [MATHEMATICAL REASONING] **Correct Option: (B)** $\frac{5}{11}$

Recurring Decimals Difference:
Convert to fractions with denominator 99:

$$0.\overline{72} = \frac{72}{99}, \quad 0.\overline{27} = \frac{27}{99}$$

Subtract:

$$\frac{72}{99} - \frac{27}{99} = \frac{45}{99}$$

Reduce by dividing numerator and denominator by 9:

$$\frac{45 \div 9}{99 \div 9} = \frac{5}{11}$$

Q35 [MATHEMATICAL REASONING] **Correct Option: (B)** $8.35 \leq x < 8.45$

Rounding Bounds and Uncertainty:
A number rounds to 8.4 to one decimal place if its hundredths digit rounds either down (from 8.40 up to strictly less than 8.45) or up (from 8.35 upwards).
Thus, the exact half-open interval of possible values is:

$$8.35 \leq x < 8.45$$

Q36 [EVERYDAY MATHEMATICS] **Correct Option: (D)** $16.0\ \text{km/L}$

Unit Rate Fuel Mileage:
Divide distance by fuel volume:

$$\text{Mileage} = \frac{456.8}{28.55} = \frac{45680}{2855}$$

Long division: $2855 \times 16 = 45680$.
Thus, the average fuel economy is exactly $16.0\ \text{km/L}$.

Q37 [EVERYDAY MATHEMATICS] **Correct Option: (A)** $\text{Rs}\ 665.50$

Commercial Decimal Billing:
1. Cost of apples:

$$3.75 \times 120.40 = 451.50\ \text{rupees}$$

2. Cost of grapes:

$$2.5 \times 85.60 = 214.00\ \text{rupees}$$

3. Total expenditure:

$$451.50 + 214.00 = \text{Rs}\ 665.50$$

Q38 [EVERYDAY MATHEMATICS] **Correct Option: (A)** $\text{Rs}\ 20,863.00$

Currency Conversion Rate:
Multiply amount in USD by exchange rate:

$$\text{Rupees} = 250 \times 83.4520 = \frac{1000}{4} \times 83.4520 = \frac{83452.0}{4} = \text{Rs}\ 20,863.00$$

Q39 [EVERYDAY MATHEMATICS] **Correct Option: (D)** $1.86\ \text{mL}$

Medical Dosage Calculation:
Multiply body weight by dosage rate:

$$\text{Dosage} = 24.8 \times 0.075 = 24.8 \times \frac{75}{1000} = \frac{1860.0}{1000} = 1.86\ \text{mL}$$

Q40 [EVERYDAY MATHEMATICS] **Correct Option: (C)** 0.084 seconds

High-Precision Time Intervals:
• Gold medalist (fastest / least time): Runner C (10.098 s)
• Bronze medalist (3rd place / greatest time): Runner B (10.182 s)
Difference:

$$10.182 - 10.098 = 0.084\ \text{seconds}$$

Q41 [EVERYDAY MATHEMATICS] **Correct Option: (B)** $4,265.7$ litres

Rainfall Volume Conversion:
1. Convert depth to metres:

$$35.4\text{ mm} = \frac{35.4}{1000} = 0.0354\text{ m}$$

2. Volume in cubic metres:

$$V = \text{Area} \times \text{depth} = 120.5 \times 0.0354 = 4.2657\text{ m}^3$$

3. Convert cubic metres to litres ($1\text{ m}^3 = 1000\text{ L}$):

$$4.2657 \times 1000 = 4,265.7\text{ litres}$$

Q42 [EVERYDAY MATHEMATICS] **Correct Option: (D)** Rs $1,744.65$

Linear Utility Billing:
1. Energy charges:

$$245.8 \times 6.75 = 1659.15\text{ rupees}$$

2. Add fixed maintenance rent:

$$1659.15 + 85.50 = \text{Rs } 1,744.65$$

Q43 [EVERYDAY MATHEMATICS] **Correct Option: (C)** 90

Measurement Division:

$$\text{Number of pieces} = \frac{157.5}{1.75} = \frac{15750}{175} = \frac{630}{7} = 90\text{ pieces}$$

Q44 [EVERYDAY MATHEMATICS] **Correct Option: (C)** 10.684 g

Decimal Purity Mass:
Multiply total medallion mass by purity fraction:

$$\text{Pure Gold} = 11.664 \times 0.916 = 10.684224\text{ g}$$

Rounding to three decimal places yields 10.684 g.

Q45 [EVERYDAY MATHEMATICS] **Correct Option: (A)** 51.15 km

Speed-Time-Distance Multiplication:

$$\text{Distance} = \text{Speed} \times \text{Time} = 18.6 \times 2.75 = 18.6 \times \frac{11}{4} = \frac{204.6}{4} = 51.15\text{ km}$$

Q46 [HOTS] **Correct Option: (B)** 0.2

Telescoping Decimal Product:
Convert each factor to its rational fractional equivalent:
- $1 - 0.5 = 1 - \frac{1}{2} = \frac{1}{2}$
- $1 - 0.\bar{3} = 1 - \frac{1}{3} = \frac{2}{3}$
- $1 - 0.25 = 1 - \frac{1}{4} = \frac{3}{4}$
- $1 - 0.2 = 1 - \frac{1}{5} = \frac{4}{5}$

Multiply together:

$$P = \frac{1}{2} \times \frac{2}{3} \times \frac{3}{4} \times \frac{4}{5}$$

All intermediate numerators and denominators cancel:

$$P = \frac{1}{5} = 0.2$$

Q47 [HOTS] **Correct Option: (C)** 8

Repetend Periodicity and Modular Arithmetic:
The repetend of $\frac{1}{7}$ consists of the 6 repeating digits:

$$(1, 4, 2, 8, 5, 7)$$

The period length is 6.
To find the digit at position 2026, divide 2026 by 6:

$$2026 = 6 \times 337 + 4 \implies 2026 \equiv 4 \pmod 6$$

The 4th digit in the repeating block $(1, 4, 2, \mathbf{8}, 5, 7)$ is strictly 8.

Q48 [HOTS] **Correct Option: (B)** 3.6288×10^{-4}

Algebraic Invariant under Decimal Transformation:
Notice that:

$$1 - (x + y - xy) = (1 - x)(1 - y)$$

Therefore, replacing (x, y) with $x + y - xy$ preserves the product of $(1 - \text{each number})$ on the chalkboard!
Let P be the invariant product:

$$P = (1 - 0.1)(1 - 0.2)(1 - 0.3)\ldots(1 - 0.9)$$

$$P = 0.9 \times 0.8 \times 0.7 \times 0.6 \times 0.5 \times 0.4 \times 0.3 \times 0.2 \times 0.1$$

$$P = \frac{9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{10^9} = \frac{9!}{10^9} = \frac{362880}{10^9} = 0.00036288$$

If F is the single final number remaining, it must satisfy:

$$1 - F = P = 0.00036288 = 3.6288 \times 10^{-4}$$

Q49 [HOTS] **Correct Option: (B)** 0.64

Discrete Optimization via AM-GM Inequality:
By the Arithmetic Mean - Geometric Mean (AM-GM) inequality, for positive real numbers x and y:

$$\sqrt{xy} \le \frac{x + y}{2}$$

Given $x + y = 1.6$:

$$\sqrt{xy} \le \frac{1.6}{2} = 0.8 \implies xy \le (0.8)^2 = 0.64$$

Equality holds when $x = y = 0.8$.
Therefore, the maximum possible product is 0.64.

Q50 [HOTS] **Correct Option: (D)** 23

Nested Recurring Continued Fraction:
1. Innermost recurring term: $0.\bar{3} = \frac{1}{3}$.
2. Reciprocal: $\frac{1}{0.\bar{3}} = \frac{1}{1/3} = 3$.
3. Denominator: $1 + 3 = 4$.
4. Fractional quotient: $\frac{1}{4}$.
5. Outer term: $0.\bar{6} = \frac{2}{3}$.
6. Full sum:

$$x = \frac{2}{3} + \frac{1}{4} = \frac{8 + 3}{12} = \frac{11}{12}$$

Since $\gcd(11, 12) = 1$, $p = 11$ and $q = 12$.

$$p + q = 11 + 12 = 23$$

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